fn sample_geometric_exp_slow

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This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of `fn sample_geometric_exp_slow` in `mod.rs` at commit 0be3ab3e6 (outdated[1]). This proof is an adaptation of subsection 5.2 of [CKS20].

1 Vetting history

- Pull Request #519

2 Pseudocode

Precondition

$\mathtt{x} \in \mathbb{Q} \wedge \mathtt{x} > 0$

Implementation

```
def sample_geometric_exp_slow(x) -> int:
    k = 0
    while True:
        if sample_bernoulli_exp(x):
            k += 1
        else:
            return k
```

Postcondition

`sample_geometric_exp_slow` returns an integer sample from $Geometric(1 - exp(-x))$, or an error propagated from `sample_bernoulli_exp`.

3 Proof

Let r be the return value of `sample_geometric_exp_slow(x)`, assuming the preconditions are met, and the function does not return an error.

Theorem 3.1. r is a realization of $R \sim Geometric(1 - exp(-x))$ [CKS20].

[1] See new changes with `git diff 0be3ab3e6..cb54ec2db rust/src/traits/samplers/cks20/mod.rs`

Proof. Let $B_i \sim Bernoulli(exp(x))$ be the sequence of booleans returned from `sample_bernoulli_exp`.

$$\begin{aligned}
P[R = k] &= P[B_1 = B_2 = ... = B_k = \bot \wedge B_{k+1} = \top] \\
&= P[B_{k+1} = \top] \prod_{i=1}^{k} P[B_i = \bot] && \text{All } B_i \text{ are independent.} \\
&= exp(-x)(1 - exp(-x))^k
\end{aligned}$$

□

References

[CKS20] Clément L. Canonne, Gautam Kamath, and Thomas Steinke. The discrete gaussian for differential privacy. *CoRR*, abs/2004.00010, 2020.